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An IoT-Integrated Autonomous Fire Extinguishing and Surveillance Robot Development Using Smart Camera Module

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***Abstract*—The Internet of Robotics, an evolving field driven by the ongoing Internet of Things (IoT) revolution, is increasingly integrating robotics into various everyday activities. This development has reached a mature stage, continually advancing with new discoveries. Robots are now capable of performing a wide range of complex tasks that were previously impossible or dangerous for humans. This research proposes the development of an IoT-enabled smart autonomous robot with surveillance capabilities to detect and extinguish fires. To achieve this, the system utilizes an Espressif Systems 32 Camera (ESP32-CAM) module, an Arduino microcontroller, and two flame sensors. Experimental results demonstrate the robot's effectiveness in patrolling busy areas. The robot is equipped with an ESP32-CAM for real-time monitoring and fire detection, a gear motor and motor driver for mobility, a flame sensor for fire detection, and a relay driver for pump control. Upon receiving a signal from the flame sensor, the Arduino initiates the extinguishing procedure. The ESP32-CAM provides real-time video, which can be monitored and controlled via a smartphone, while the flame sensor continuously scans the environment. Upon fire detection, the system activates the water pump, spraying water through a sprinkler until the fire is extinguished.**

***Index Terms*—IoT, Esp32-CAM, Flame Sensor, Autonomous Robot, Arduino UNO, Fire Detection, Data Analysis.**

I. Introduction

Surveillance activities, both indoors and outdoors, can be carried out by humans or with the assistance of embedded systems such as robots and automated devices. Robots, which are machines that operate autonomously, replace human efforts in various tasks [1], [2]. The integration of autonomous robotic systems in fire safety has garnered significant attention due to their ability to operate in hazardous environments where human intervention is challenging. However, most existing fire-fighting robots focus primarily on either fire detection or suppression, without offering a comprehensive, cost-effective, and remotely accessible solution.

In this paper, we propose a novel IoT-enabled fire-fighting robot that combines real-time fire detection, autonomous navigation, and remote-controlled extinguishing mechanisms. Unlike traditional systems that rely on expensive sensors and complex control mechanisms, our approach uses an ESP32-CAM module for real-time surveillance and an Arduino UNO for cost-effective decision-making, making it highly suitable for resource-constrained environments. Moreover, the rapid response of firefighters can significantly mitigate damage and reduce fatalities in fire incidents. With technological advancements, the gap between firefighting and automation has been bridged, leading to more effective and efficient fire-extinguishing methods. Over the years, the incidence of fire-related accidents in Bangladesh and other regions has surged, particularly in domestic buildings, resulting in devastating consequences such as asset losses, injuries, and fatalities. The increasing number of deaths from fire accidents, especially in residential areas, underscores the urgent need for improved fire safety measures [3]. Common causes of household fires include cooking, smoking, and the use of candles. The primary objective of this project is to contribute to the advancement of automation systems by developing an autonomous fire-extinguishing robot. To address these challenges, our system integrates a dual-mode operation: one for autonomous fire detection and suppression, and another for manual intervention in complex scenarios. This dual functionality enhances adaptability, making it an efficient and versatile solution for real-world fire safety applications.

Fire spreads rapidly, with the potential to double in size every minute, leading to a rapid accumulation of smoke, a major cause of fatalities. Therefore, early fire detection is crucial, as it significantly improves survival chances. The key novelty of our work lies in its cost-effectiveness, seamless IoT connectivity for remote monitoring, and a hybrid control approach, ensuring robust fire mitigation even in unpredictable fire situations. The main contributions of this paper are as follows:

- We develop an IoT-enabled fire-extinguishing robot with dual functionality: autonomous fire detection and remote-controlled operation.
- The system integrates an ESP32-CAM module for real-time video surveillance and an Arduino UNO for efficient, low-cost processing, ensuring affordability without compromising performance.
- We implement a novel fire detection and suppression strategy utilizing sensor fusion and real-time image processing, improving accuracy and response time.
- Our robot features IoT-based remote monitoring, allowing users to control fire suppression remotely via a smartphone or web interface, enhancing accessibility and usability.
- We conduct extensive real-world experiments to validate the system's efficiency, demonstrating its capability to operate effectively in hazardous, fire-prone environments.

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TABLE I
COMPARISON OF IOT-BASED AUTONOMOUS FIRE EXTINGUISHING SYSTEMS

Study	Technologies Used	Methods	Key Features	Strengths	Weaknesses
Irshad et al. (2024) [4]	Arduino, Fire Sensor, Smoke Sensor	Water Pumping	Smoke Sensors, Water Pump, Servo Motor	Autonomous navigation, fire detection, water extinguishing	Limited fire response range, sensor sensitivity issues
Rao et al. (2024) [5]	ThingSpeak, IoT, Smoke, Temperature, Gas Sensors	IoT-Based Actuator Control	Real-time fire detection, actuator control, operates in hazardous environments	Real-time data transmission, remote access, multi-sensor integration	Dependency on internet, maintenance of IoT platform
Kanwar et al. (2024) [6]	Arduino, IoT, Fire-Type Sensors, Carbon Monoxide Sensors	Notification System	Remote fire detection, authority notification, gas level analysis	Provides real-time updates, assists in assessing risks	Limited coverage area, dependency on network reliability
Diwanji et al. (2019) [7]	Arduino Uno, DC Motors, Sensors	Water Spraying	Sensor-based movement control, fire detection, water spraying mechanism	Efficient hardware integration, fire suppression accuracy	Limited obstacle detection, setup complexity
Arora et al. (2024) [8]	Infrared Flame Sensors, Ultrasonic Sensors, Arduino Uno	Water Spraying	Multi-directional fire detection, obstacle navigation, water spraying	Efficient multi-directional detection, effective obstacle avoidance	Limited response time, expensive components
Rehman et al. (2024) [9]	Heat Sensors, Servo Motors, Water Gun, Arduino Uno	Targeted Water Spraying	Early fire detection, targeted water spraying, scans heat gradients	Quick suppression of fires, heat detection accuracy	Requires accurate calibration, limited range of detection
Sures et al. (2024) [10]	Flame Detection Sensors, CO2 Extinguishing, Servo Motors	Carbon Dioxide Suppression	Carbon dioxide-based extinguishing, flame detection	Immediate response to indoor fires, minimizes human risk	Not suitable for all types of fires, setup complexity
Proposed System	ESP32-CAM, Arduino Uno, Flame Sensors, Water Pump, Servo Motors, Buzzer, Mobile App	Water Extinguishing, Remote Monitoring	Fire detection, water extinguishing, real-time monitoring, mobile control	Integrated fire detection and suppression, remote operation	Dependence on WiFi, setup complexity, real-time latency issues

II. LITERATURE REVIEW

In this section, we present a review of literature on IoT-based systems for autonomous fire extinguishing, as well as development strategies for efficient smart surveillance robotic technologies.

For instance, Irshad et al. [4] designed Arduino-based firefighter robots that leverage open-source electronics for user-friendly operation. These robots focus on autonomous navigation and fire detection using fire and smoke sensors. Upon fire detection, they activate a water-pumping motor to extinguish the flames, aiming to reduce risks for human firefighters, improve situational awareness, and optimize resource usage during emergencies. Similarly, another study [5] highlighted an IoT-enabled fire-fighting robot integrated with the ThingSpeak platform. Equipped with temperature, smoke, and gas sensors, the robot autonomously detects and responds to fire outbreaks in real-time. The robot uses onboard actuators and fire suppression mechanisms, mitigating damages and minimizing risks to human life. The system's ability to operate in hazardous environments, such as chemical factories, underscores its critical role in modern firefighting practices. Furthermore, [6] discussed advancements in robotics technology, emphasizing that robots are increasingly replacing humans in various tasks, including firefighting, to prevent frequent fire accidents that endanger lives and property. This IoT-based firefighting robot detects fires and notifies authorities, enabling remote visualization of fire locations and communication with affected individuals via an automatic receiver. Equipped with sensors, it also assesses fire type and carbon monoxide levels, generating analytical graphs to aid safety authorities in evaluating gas exposure and mitigating harm.

On the other hand, Diwanji et al. [7] explored the integration of automation into firefighting, emphasizing the role of robots in preemptively locating and controlling fires to enhance safety. The fire-fighting robot, detailed in this study, uses DC motors and sensors interfaced with an Arduino Uno board. Programming in the Arduino IDE facilitates sensor-based movement control, which is crucial for effective fire suppression. This approach aims to collaborate with firefighters to mitigate risks and reduce injuries in hazardous environments. [8] discusses the development of an autonomous firefighting robot designed to enhance safety and efficiency in fire suppression. Equipped with infrared flame sensors, microcontrollers, and ultrasonic sensors, the robot detects fires in multiple directions and navigates around obstacles. It features a sophisticated water-spray extinguishing mechanism controlled by an Arduino Uno, ensuring prompt and effective responses to both electrical and conventional fires. In similar work, [9] introduced an autonomous fire extinguishing system designed to address the destructive impact of fires on life, property, and the environment. By targeting the initial stages of fire ignition using heat sensors, a flow control system, servo motors, and a water extinguishing gun, the system aims to suppress fires before they escalate. The paper evaluates the system's efficiency, suppression accuracy, response time, power consumption, and coverage area, proposing future enhancements to improve its design and effectiveness. Lastly, [10] aimed to develop an autonomous robot equipped with flame detection sensors and carbon dioxide extinguishing capabilities, reducing the risk to human life by promptly addressing indoor fire incidents through robotic intervention.

In summary, various studies have proposed systems for fire control using robotics technology in smart environments. Table I summarizes these literature works. Based on the insights gained from these studies, we propose an administration remote-controlled robot utilizing the ESP32-CAM and Arduino Uno module. Additionally, the system emphasizes prompt fire detection and extinguishing abilities to enhance security and efficiency.

III. PROPOSED METHODOLOGY FOR IOT-BASED AUTONOMOUS FIRE EXTINGUISHING AND SURVEILLANCE ROBOT SYSTEM

In this paper, we focused on enhancing a robot car equipped with a left-hand drive system, introducing significant modifications to its RF and Arduino circuits to achieve advanced functionality. The car is equipped with both front and rear motors that facilitate its directional control: the front-wheel drive is used for steering, allowing the car to turn left or right, while the rear-wheel drive controls the car's forward and backward movements. Central to the system's operation is the ESP32-CAM module, which manages the car's movements, video streaming, and WiFi communication. The motor driver, connected to the ESP32-CAM, controls the power supplied to the motors, thereby managing the car's directional movements. Commands are sent from an Android phone via WiFi, allowing

for wireless remote control. The phone communicates with the ESP32-CAM module to execute movement commands.

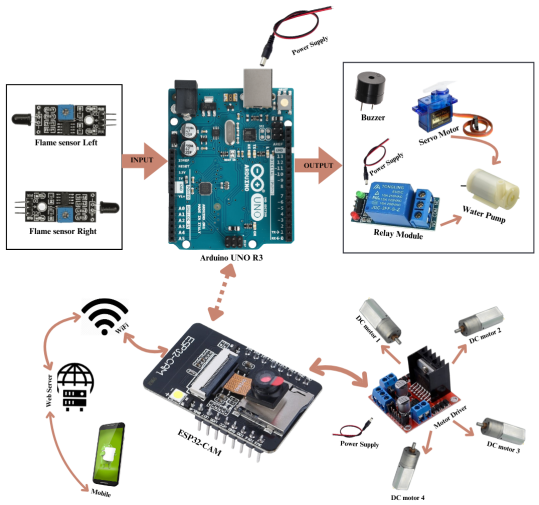


Fig. 1. Proposed Block Diagram for Controlling each Connection

However, Fig. 1 shows the block diagram for controlling each connection, starting with the command input on the Android device and going all the way to the autonomous fire detection and suppression system with the flame sensors, water pump, and servo motor. It also includes motor control for movement, light control for visibility, and processing from the ESP32-CAM module.

A. Fire Detection Algorithm & Movement Control

The system integrates flame sensors, strategically placed for optimal coverage, and real-time video stream analysis to detect heat signatures while filtering out false positives caused by sunlight or other heat sources. A decision-making mechanism processes data from multiple sensors, dynamically adjusting sensitivity to account for environmental factors, ensuring precise and reliable fire detection [1]. Once a fire is detected, the robot calculates the shortest route using the A* pathfinding algorithm while avoiding obstacles detected by ultrasonic sensors. A Proportional-Integral-Derivative (PID) controller adjusts the robot's speed to ensure smooth and adaptive navigation in dynamic environments, enabling efficient and accurate operation [2].

B. Analysis of System Design

The two main components of the system we designed are the surveillance and fire extinguishing parts. These portions provide smooth mobile WiFi connectivity between laptops, mobile devices, and the robot. Real-time observation and fire detection and extinguishment comprise the robot's functionalities. The automatic extinguishing process begins as soon as the flame sensors detect a fire. To put out the fire, the robot has actuators on board, such as a servo motor and a water pump. An Arduino microcontroller powers the water pump, which squirts water over the flames. The water spray is precisely targeted by the servo motor, which also modifies its angle and direction as necessary. In the meantime, a buzzer sounds a warning to consumers that there is a fire. In Algorithm 1, the pseudocode represents the logic for the Autonomous Fire Extinguishing and Surveillance Robot.

The servo motor control that adjusts the water spray can be modeled as a function of the time-dependent angle $\theta(t)$ and its angular velocity ω, as described in equation 1. This can be expressed as:

Algorithm 1 IoT-enabled Autonomous Fire Extinguishing and Surveillance Robot

Initialize: $\{M, E, W, S, L, P, A, F_L, F_R\}$ ▷ Motor, ESP32-CAM, WiFi, Servo, Light, Pump, Arduino, Left Flame, Right Flame
procedure CAR_CONTROL
  $W \leftarrow 1$ ▷ WiFi on
  $V \leftarrow 1$ ▷ Video streaming on
  $C \leftarrow$ User Command(M) ▷ User control: movement
  $L \leftarrow$ User Command(L) ▷ Set headlight intensity
  $M_s \leftarrow$ User Command(M) ▷ Set speed
  while $V = 1$ **do**
    $M_{act} \leftarrow f(M_s)$ ▷ Move car with speed M_s
    $L_{int} \leftarrow f(L)$ ▷ Set light intensity
  end while
end procedure
procedure FIRE_DETECTION_AND_EXTINGUISH
  $F_L \leftarrow$ Flame Sensor(A_L) ▷ Left flame sensor
  $F_R \leftarrow$ Flame Sensor(A_R) ▷ Right flame sensor
  if $F_L = 1 \wedge F_R = 0$ **then**
    $C_L \leftarrow 1$ ▷ Move left
    $B \leftarrow 1$ ▷ Buzzer on
    $P \leftarrow 1$ ▷ Activate pump
    $S_L \leftarrow f(P)$ ▷ Servo to left
  else if $F_R = 1 \wedge F_L = 0$ **then**
    $C_R \leftarrow 1$ ▷ Move right
    $B \leftarrow 1$ ▷ Buzzer on
    $P \leftarrow 1$ ▷ Activate pump
    $S_R \leftarrow f(P)$ ▷ Servo to right
  else if $F_L = 1 \wedge F_R = 1$ **then**
    $C_{LR} \leftarrow 1$ ▷ Move left-right
    $B \leftarrow 1$ ▷ Buzzer on
    $P \leftarrow 1$ ▷ Activate pump
    $S_{LR} \leftarrow f(P)$ ▷ Servo between left and right
  end if
end procedure
procedure SYSTEM_OPERATION
  While system on:
    $C \leftarrow$ Car_Control ▷ Car movement & lights
    $F_{ext} \leftarrow$ Fire_Detection_and_Extinguish ▷ Fire detection
    $S_{monitor} \leftarrow f(V)$ ▷ Monitor video feed
  if $F_L = 0 \wedge F_R = 0$ **then**
    $P_{off} \leftarrow 0$ ▷ Deactivate pump if no fire
  end if
end procedure

$$\theta(t) = \theta_{\text{start}} + \Delta\theta \cdot \sin(\omega t) \tag{1}$$

where $\theta(t)$ is the angle of the servo motor at time t, θ_{start} is the starting angle of the servo motor, $\Delta\theta$ is the amplitude of the angle change, and ω is the angular velocity. To ensure accurate targeting of the fire, the system integrates a thermal sensor and an ESP32-CAM module for real-time fire detection. The fire's position is determined based on the centroid of the detected flame in the captured image. The servo motor then adjusts its angle $\theta(t)$ dynamically to align the water nozzle with the fire source. The detection and targeting mechanism operates within an effective range of 1 to 1.5 meters, ensuring precise fire suppression without excessive water dispersion. The system continuously refines the servo motor angle based on updated fire position data, improving accuracy even if the fire spreads or moves. This adaptive approach enhances firefighting efficiency and minimizes water wastage.

The water pump is activated when a fire is detected. The control signal $u(t)$ for the water pump can be described as a binary function, as shown in equation 2. This can be represented as:

$$u(t) = \begin{cases} 1 & \text{if fire is detected, i.e., } F(t) = 1 \\ 0 & \text{if no fire is detected, i.e., } F(t) = 0 \end{cases} \tag{2}$$

where $u(t)$ is the control signal for the water pump, and $F(t)$ is the fire detection signal from the flame sensor.

The detection of fire is based on the flame sensor's infrared intensity, $I(t)$, and is thresholded to determine fire presence as shown in equation 3. The fire detection logic is given by:

$$F(t) = \begin{cases} 1 & \text{if } I(t) \geq I_{\text{threshold}} \\ 0 & \text{if } I(t) < I_{\text{threshold}} \end{cases} \tag{3}$$

where $I(t)$ is the intensity of the infrared signal at time t, and $I_{\text{threshold}}$ is the threshold intensity for fire detection.

The robot's movement control is based on its velocity components $v_x(t)$ and $v_y(t)$ in the x- and y-directions. The position coordinates $x(t)$ and $y(t)$ of the robot are determined by integrating the velocity over time as shown in equations 4 and 5. These are given by:

$$x(t) = x_0 + \int_0^t v_x(t')dt' \tag{4}$$

$$y(t) = y_0 + \int_0^t v_y(t')dt' \tag{5}$$

where $x(t)$, $y(t)$ are the position coordinates at time t, x_0, y_0 are the initial coordinates at time $t = 0$, and $v_x(t)$, $v_y(t)$ are the velocity components in the x- and y-directions at time t.

The communication between the robot and the mobile interface occurs over a WiFi connection. The data transmission rate $R(t)$ can be modeled as shown in equation 6. The rate is given by:

$$R(t) = \frac{B}{T(t)} \tag{6}$$

where B is the available bandwidth of the WiFi network, and $T(t)$ is the transmission time for data at time t.

Once a fire is detected, the system responds by activating the water pump and targeting the flame with the servo motor. The robot's behavior can be modeled by a feedback control system that modifies its trajectory or responds to changing conditions based on sensor input. For example, the robot's velocity components $v_x(t)$ and $v_y(t)$ are adjusted depending on whether a fire is detected or if the robot is navigating an obstacle. The general form for modifying the robot's movement based on sensor feedback is shown in equations 7 and 8. These adjustments are given by:

$$v_x(t) = v_x(t) + \alpha \cdot F(t) \tag{7}$$

$$v_y(t) = v_y(t) + \beta \cdot F(t) \tag{8}$$

where $v_x(t)$, $v_y(t)$ are the velocity components in the x- and y-directions, and α, β are constants that define how much the velocity is adjusted in response to fire detection.

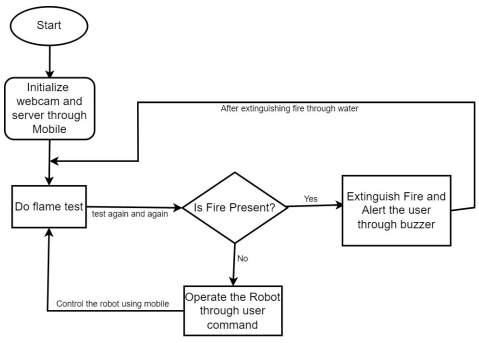


Fig. 2. Data Flow Diagram of Proposed System

Fig. 2 illustrates the system starting the webcam and server. Upon initialization, the system starts the webcam and server through the mobile interface. The flame sensors continuously monitor for fires. If a fire is detected, the extinguishing process is activated, involving the water pump and servo motor, while the buzzer alerts the user. Once the fire is extinguished, the system resumes monitoring. If no fire is detected, the robot proceeds with its programmed movements. Throughout these processes, the user can control and oversee the robot's operations via a mobile device.

C. Collecting of Hardware Requirements

This robot requires several essential hardware components for its proper functioning. Table II summarizes the specific roles, functionalities, and specifications of each hardware component used in this system. The hardware components were selected based on their availability, performance, and compatibility with the overall design. Each component plays a crucial role in the robot's autonomous operation and real-time response to fire hazards.

TABLE II
FUNCTIONALITIES OF HARDWARE COMPONENTS IN THE ROBOT

Component	Functionality	Specifications
Arduino UNO	Controls the actuators (servo motor, water pump) and processes input from the flame sensor. Interfaces with other components like the buzzer for alerts.	ATmega328P microcontroller, 16 MHz clock speed, 6 analog inputs, 14 digital I/O pins, USB connection, 5V operating voltage. [11]
Motor Driver	Amplifies low-current control signals from Arduino to drive the motors that control the robot's movement.	L298N dual H-bridge, 7-35V input voltage, 2A current per channel, built-in 5V regulator. [12]
Flame Sensor	Detects infrared light from fire. Provides input to the system to trigger the extinguishing process.	KY-026 model, adjustable sensitivity via potentiometer, analog and digital outputs. [13]
Buzzer	Provides an audible alert when a fire is detected. Activated by the Arduino when a fire is identified.	5V, piezoelectric buzzer, frequency range: 2-5 kHz, 85 dB sound output. [14]
ESP32-CAM	Provides real-time video streaming for surveillance. Enables WiFi communication between the robot and the mobile interface.	2MP OV2640 camera, Wi-Fi and Bluetooth support, 80-240 MHz clock speed, 6mA deep sleep current. [15]
Servo Motor	Controls the direction and targeting of the water spray to put out the fire. Adjusts based on the Arduino control signals.	180° rotation, 4.8-6V operating voltage, torque: 2.5-3.5 kg/cm, 0.12-0.14 sec/60° speed. [16]
Mini Water Pump	Pumps water for extinguishing the fire. Activated by Arduino based on the fire detection signal.	3-6V operating voltage, 220mA current draw, 120 L/h flow rate, submersible design. [17]
Chassis	Provides the physical structure for the robot and enabling movement. Equipped with DC motors for locomotion.	2-layer acrylic material, 3-12V rated DC motors, 120 RPM speed, 800 gf-cm torque, 65mm wheels. [18]

D. Cost Analysis

The proposed system is cost-effective, utilizing affordable components such as the Arduino UNO and ESP32-CAM for efficient performance while minimizing costs. The selection of a mini water pump and servo motor reduces power consumption. Compared to traditional fire-fighting robots with complex hardware, our system offers a reliable, low-cost alternative ideal for resource-constrained environments. The cost breakdown of the hardware components is presented in Table III.

E. Arrangement of Software Requirements

1) Arduino IDE: The Arduino IDE is an open-source software which is used to write and upload code to the Arduino boards [19]. The IDE is compatible with Windows, macOS, and Linux, and supports C and C++ programming. It's an Integrated Development Environment (IDE) where we write and upload code, known as sketches, to boards like the ESP-32 Camera Module and Arduino UNO.

TABLE III
COST ANALYSIS OF HARDWARE COMPONENTS IN THE ROBOT

Component	Functionality	Unit Price (USD)	Total Cost (USD)
Arduino UNO	Controls actuators, processes sensor input, interfaces with other components.	10	10
Motor Driver	Amplifies control signals to drive motors.	5	5
Flame Sensor	Detects infrared light from fire.	3	6
Buzzer	Provides an audible alert for fire detection.	2	2
ESP32-CAM	Provides real-time video streaming and WiFi communication.	15	15
Servo Motor	Controls the water spray direction.	8	8
Mini Water Pump	Pumps water for fire extinguishing.	7	7
Chassis	Provides the physical structure and enables movement.	12	12
Total Cost			**65**

2) Web Browser: Web browser interface enhances user interaction significantly. Through the interface, users can monitor real-time data such as controlling the system process.

IV. RESULT ANALYSIS AND PERFORMANCE MEASUREMENT

A. Flame Sensor Behavior Under Different Fire Conditions

Fig. 3(a) shows the flame sensor readings over a 24-hour period under two conditions: with a fire source and without a fire source. The x-axis represents the time of day in hours, and the y-axis represents the sensor's analog output (AO) readings. The gray line with squares shows the readings when a fire source was present, starting lower but steadily increasing and stabilizing at a higher value, indicating the presence of fire. In contrast, the green line with squares, representing the readings without a fire source, starts higher but fluctuates significantly, showing more variability and lower average readings. This pattern demonstrates the sensor's ability to detect fire, with higher and more consistent readings when a fire is present, and lower, more unstable readings when no fire is detected.This analysis focuses on the sensor's behavior under different fire conditions. To better support the discussion, more precise labeling and clearer figures have been provided.

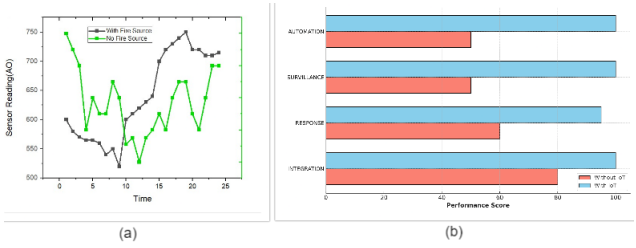


Fig. 3. (a) Effect of Sensitivity of Flame Sensor, (b) Comparison of the proposed system with non-IoT System

B. Comparison with IoT and Non-IoT System

Fig. 3(b) illustrates the performance differences across four aspects—Integration, Response, Surveillance, and Automation—when utilizing IoT technology versus not using it. Performance levels are shown on a scale from 0 to 100. The graph reveals that IoT significantly enhances performance in all aspects. Integration and Surveillance both reach a perfect score of 100 with IoT, compared to slightly above 60 without it. Automation shows the most dramatic improvement, soaring from around 40 without IoT to 100 with IoT. The response also sees a notable increase, improving from around 60 to slightly above 80 with IoT. Overall, the graph underscores the substantial benefits of implementing IoT technology in enhancing these critical performance areas.

C. Real-Time Analysis

The IoT-enabled autonomous firefighting robot we developed showcased robust functionality and impressive performance during various experimental tests. Initially, we powered on the robot and established a Wi-Fi connection using the ESP32-CAM module. This setup allowed us to access the robot's control interface and real-time video stream directly through a mobile browser, enabling remote monitoring and operation. As shown in Fig. 4(a), the mobile interface provided intuitive controls for maneuvering the robot and viewing live video feeds from the ESP32-CAM. This made remote management of the robot's actions seamless and user-friendly. Fig. 4(b) further illustrates the integrated controller interface alongside the real-time video stream, highlighting the comprehensive monitoring and control capabilities available from a mobile device. The robot's firefighting capabilities were put to the test using dual fire sensors positioned on the left and right sides of the robot. When a fire was detected, an alert was triggered, activating a buzzer and a water pump controlled by a servo motor, as depicted in Fig. 4(c).

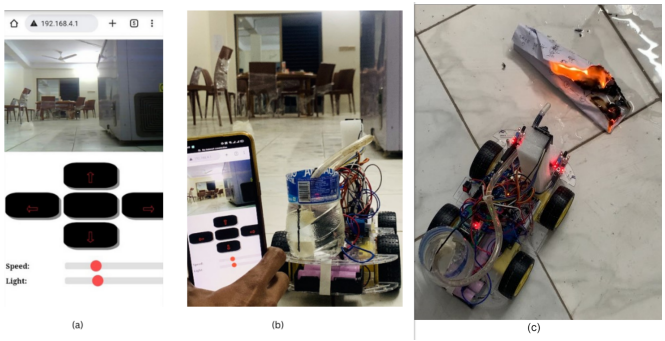


Fig. 4. (a) Controller on mobile phone screen, (b) Controller and real-time Video on Mobile Phone Screen, (c) Fire Extinguisher Mode

This process was streamed live to a mobile screen, providing real-time monitoring and situational awareness. Our experimental results demonstrate the successful implementation and operation of the Surveillance Firefighting Robot. It effectively achieved the project's objectives by integrating advanced robotics, IoT technologies, and firefighting functionalities. The robot's ability to detect, respond to, and extinguish fires autonomously underscores its potential to enhance emergency response capabilities and safeguard environments from fire-related risks.

D. Challenges in Real-World Deployment

While the robot performed well in controlled environments, several challenges arise in real-world deployment. Key obstacles include battery life, sensor interference from rain, smoke, dust, and the need for regular maintenance. Environmental factors such as extreme temperatures, high humidity, and varied topography can reduce system performance and reliability. Sensor accuracy, especially in flame detection, is crucial for timely fire identification but may be affected by calibration issues, rain, dust, or extreme temperatures. Heavy rain or

water exposure can interfere with sensor sensitivity, leading to incorrect readings or malfunctions, while dust accumulation may obstruct the sensors, requiring regular maintenance. Extreme temperatures can also degrade components over time, reducing their lifespan and affecting detection capabilities. These environmental conditions must be considered and mitigated to ensure the system's reliability in dynamic, real-world environments.

E. Security and Robustness Considerations

To enhance the system's reliability in real-world conditions, we addressed potential challenges like network failure and power outages. In case of network failure, the robot continued to operate autonomously with local failover systems that allowed basic functions like fire suppression or returning to a safe location. For power outages, backup power sources such as rechargeable batteries ensured uninterrupted operation. Additionally, power-saving modes extended battery life. To secure the system, we implemented end-to-end encryption and secure authentication protocols to protect communication channels and sensitive data. Regular security audits and software updates also helped safeguard against cyber threats. These measures improved the system's robustness and security, ensuring reliable performance even under adverse conditions.

F. Applications

Our IoT-enabled autonomous fire extinguishing and surveillance robot enhances fire safety by improving response times, safety, and efficiency across various industries. In industrial settings like factories and warehouses, it provides continuous monitoring, minimizing damage and swiftly detecting and extinguishing fires, ensuring a safer environment. Early fire detection reduces property damage, protects inhabitants, and enhances community safety. The robot's autonomous navigation and firefighting capabilities are particularly valuable in remote or hazardous areas, such as forests or mining sites, where human access is limited. It ensures personnel safety and protects natural resources by reaching and extinguishing fires in dangerous locations. In emergencies, the robot assists rescuers by identifying and putting out fires early, enhancing disaster resilience and ensuring efficient fire management, especially in high-risk situations.

G. Discussion

In this section, we compare various IoT-based autonomous fire extinguishing systems with our proposed model in Table I. The evaluation is based on different criteria, such as sensor technology, method of fire suppression, efficiency, and real-time monitoring capabilities. The proposed model enhances fire detection and suppression through ESP32-CAM for real-time monitoring, which is absent in other models. Additionally, the system integrates a mobile application, allowing users to control and monitor the fire extinguishing process remotely. This ensures prompt response and effective fire management compared to conventional IoT-based fire suppression systems.

V. Conclusion

In this paper, we develop an IoT-enabled wireless video surveillance fire extinguisher robot that can be controlled by an Android mobile device. Using a mobile app, web server, and Wi-Fi, this autonomous robot effectively tackles small-scale fires, meeting our project goals. This work highlights how IoT can address challenges in limited-range surveillance and firefighting. The robot can be manually controlled via a smartphone, offers automatic monitoring, and can reach hard-to-access areas due to its compact size. While it currently handles only small fires, future improvements will focus on scalability and integrating advanced sensors to expand its capabilities.

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